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Summary: Lossless Medium Propagation

Helmholtz's Equation for Propagating EM Wave

$$\nabla^2 \mathbf{E} + \mu \epsilon \omega^2 \mathbf{E} = \nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0$$

where we define $k = \omega \sqrt{\mu \epsilon}$

Wave Parameters

$$E(z, t) = A \cos(\omega t - kz) \Leftrightarrow A e^{-jkz}$$

$$\omega = 2\pi f = \frac{2\pi}{T} \quad (\text{rad/s})$$

$$k = \beta = \frac{2\pi}{\lambda} \quad (\text{rad/m})$$

u_p = phase velocity

$$u_p = \frac{\omega}{k} = \frac{c}{\sqrt{\mu_r \epsilon_r}} = f\lambda \quad (\text{m/s})$$

- velocity of wave must give direction

Ae^{-jkz} is a wave propagating in the +z direction.

Be^{+jkz} is a wave propagating in the -z direction.

u_g = group velocity (m/s)

$$u_g = \frac{\partial \omega}{\partial k} \leq c_{\text{in medium}} \quad (\text{m/s})$$

$$u_p u_g = c^2 \quad \text{or} \quad u_p u_g = c_{\text{in medium}}^2$$

group velocity is the velocity of the envelope/energy/information

phase velocity is the velocity of a wavefront/single peak

In Free Space

$$k_0 = \omega \sqrt{\mu_0 \epsilon_0} = \frac{2\pi f}{c} = \frac{2\pi}{\lambda_0}$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\lambda_0 = \text{free space wavelength} = \frac{c}{f}$$

In Arbitrary Medium with $\epsilon = \epsilon_r \epsilon_0$ and $\mu = \mu_r \mu_0$

$$k = \omega \sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0} = \omega \sqrt{\mu \epsilon} = \frac{2\pi f}{c} \sqrt{\mu_r \epsilon_r}$$

$$k = \frac{2\pi}{\lambda_0} \sqrt{\mu_r \epsilon_r} = \beta = k_0 n$$

λ = wavelength in the medium

$$\lambda = \frac{2\pi}{k} = \frac{\lambda_0}{\sqrt{\mu_r \epsilon_r}}$$

n = refractive index of medium

$$n = \frac{c}{u_p} = \frac{\sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0}}{\sqrt{\mu_0 \epsilon_0}} = \sqrt{\mu_r \epsilon_r}$$

General Instantaneous and Phasor Form

Instantaneous: $E(r, t) = a_E |E_0| \cos(\omega t - k \cdot r + \phi)$

Phasor: $E(r) = a_E |E_0| e^{j\phi} e^{-jk \cdot r} = E_0 e^{-jk \cdot r} \quad E_0 = a_E |E_0| e^{j\phi}$

- Varies in the a_E direction and propagate in the a_k direction
- Take real of phasor to get back time domain

Instantaneous: $H(r, t) = a_H |H_0| \cos(\omega t - k \cdot r + \phi)$

Phasor: $H(r) = a_H |H_0| e^{j\phi} e^{-jk \cdot r} = H_0 e^{-jk \cdot r} \quad H_0 = a_H |H_0| e^{j\phi}$

- Varies in the a_H direction and propagate in the a_k direction
- Take real of phasor to get back time domain

k = vector propagation constant = $k_x a_x + k_y a_y + k_z a_z \quad |k|$ = wavenumber

r = observation position vector = $xa_x + ya_y + za_z$

Conversion and Relationship Between E and H

$$E(r) = \eta H(r) \times a_k$$

$$E(r, t) = \eta H(r, t) \times a_k$$

$$H(r) = \frac{1}{\eta} a_k \times E(r)$$

$$H(r, t) = \frac{1}{\eta} a_k \times E(r, t)$$

$$k = |k|a_k = j\nabla$$

$$k \times E(r) = \omega\mu H(r) \quad k \cdot E(r) = 0$$

$$k \times H(r) = -\omega\epsilon E(r) \quad k \cdot H(r) = 0$$

$\eta = \text{Intrinsic impedance}$

$$\eta = \frac{|E|}{|H|} = \frac{\omega\mu}{k} = \sqrt{\frac{\mu}{\epsilon}} = \eta_0 \sqrt{\frac{\mu_r}{\epsilon_r}}$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi = 377\Omega$$

Poynting Vector

$$S(r, t) = E(r, t) \times H(r, t)$$

$$S(r, t) = \text{Re}\{E(r)e^{j\omega t}\} \times \text{Re}\{H(r)e^{j\omega t}\}$$

- This is cross product between two vectors

$$S_{ave} = \frac{1}{2} \text{Re}\{E(r) \times H(r)^*\}$$

- this is multiplication of E phasor with conjugate of H phasor

$$S_{ave} = \frac{1}{2\eta} |E_0|^2 a_k = \frac{\eta}{2} |H_0|^2 a_k = P_{ave} a_k$$

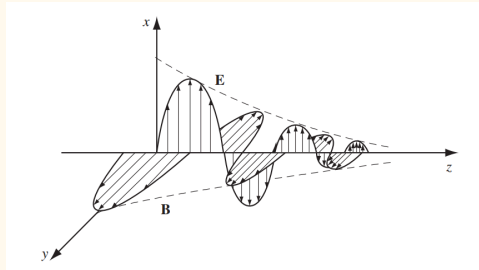
all in W/m²

S_{ave} also called power density

Summary: Lossy Medium Propagation

<u>Ampere's Law with Conduction Current</u>	<u>Complex Permittivity (F/m)</u>	<u>Loss Tangent</u>
$\nabla \times H = J + j\omega D$ $\nabla \times H = \sigma E + j\omega \epsilon E$ $\nabla \times H = j\omega(\epsilon - j\frac{\sigma}{\omega})E$ $\nabla \times H = j\omega \epsilon_c E$ σE = conduction current density $j\omega \epsilon E$ = displacement current density Out of phase by $\pi/2$	<u>Complex Permittivity</u> $\epsilon_c = \epsilon - j\frac{\sigma}{\omega}$ $\epsilon_c = \epsilon' - j\epsilon''$ $\epsilon' = \epsilon, \epsilon'' = \frac{\sigma}{\omega}$ <u>Complex Relative Permittivity</u> $\epsilon_{rc} = \frac{\epsilon_c}{\epsilon_0} = \epsilon_r - j\frac{\sigma}{\omega \epsilon_0}$ $\epsilon_{rc} = \frac{\epsilon_c}{\epsilon_0} = \epsilon_r' - j\epsilon_r''$ $\epsilon_r' = \epsilon_r, \epsilon_r'' = \frac{\sigma}{\omega \epsilon_0}$	<i>Loss tangent</i> $= \frac{\epsilon''}{\epsilon'} = \frac{\epsilon_r''}{\epsilon_r'}$ $= \frac{\frac{\sigma}{\omega}}{\epsilon} = \frac{\sigma}{\omega \epsilon}$

Helmholtz's Equation for Propagating EM Wave in Lossy Medium

<u>Complex Vector Propagation Constant</u>	<u>Complex Intrinsic Impedance</u>
$ k_c = \omega \sqrt{\mu \epsilon_c} = \omega \sqrt{\mu \epsilon_0 \epsilon_{rc}}$ $ k_c = \omega \sqrt{\mu \epsilon_0 (\epsilon_r' - j\epsilon_r'')}$ $ k_c = \beta - j\alpha \quad k_c = k_c a_k$ α = attenuation constant > 0 β = propagation/phase constant > 0 λ = wavelength in the medium $= \frac{2\pi}{\beta}$ u_p = phase velocity $= \frac{\omega}{\beta} = f\lambda$ velocity of wave must give direction	 $\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \eta_c e^{j\theta_\eta}$ $ \eta_c = \frac{ E }{ H }$ $\theta_\eta = \angle \frac{E}{H} \Rightarrow E \text{ leads } H \text{ by } \theta_\eta$

General Instantaneous and Phasor Form

Instantaneous: $E(r, t) = a_E |E_0| e^{-\alpha(a_k \cdot r)} \cos(\omega t - \beta(a_k \cdot r) + \phi)$

Phasor: $E(r) = a_E |E_0| e^{j\phi} e^{-jk_c \cdot r} = a_E |E_0| e^{j\phi} e^{-\alpha(a_k \cdot r)} e^{-j\beta(a_k \cdot r)} = E_0 e^{-\alpha(a_k \cdot r)} e^{-j\beta(a_k \cdot r)}$

$E_0 = a_E |E_0| e^{j\phi}$

- Varies in the a_E direction, propagate and attenuate in the a_k direction
- Must be in cosine
- in V/m

Instantaneous: $H(r, t) = a_H |H_0| e^{-\alpha(a_k \cdot r)} \cos(\omega t - \beta(a_k \cdot r) + \phi)$

Phasor: $H(r) = a_H |H_0| e^{j\phi} e^{-jk_c \cdot r} = a_H |H_0| e^{j\phi} e^{-\alpha(a_k \cdot r)} e^{-j\beta(a_k \cdot r)} = H_0 e^{-\alpha(a_k \cdot r)} e^{-j\beta(a_k \cdot r)}$

$H_0 = a_H |H_0| e^{j\phi} = a_H \frac{|E_0|}{|\eta_c|} e^{-j\theta_\eta} e^{j\phi}$

- Varies in the a_H direction, propagate and attenuate in the a_k direction
- Must be in cosine
- in A/m

$r = \text{observation position vector} = x a_x + y a_y + z a_z$

$E(r) = \eta_c H(r) \times a_k \quad E(r, t) = \eta_c H(r, t) \times a_k$

$H(r) = \frac{1}{\eta_c} a_k \times E(r) \quad H(r, t) = \frac{1}{\eta_c} a_k \times E(r, t)$

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-loss Medium ($\epsilon''/\epsilon' \ll 1$)	Good Conductor ($\epsilon''/\epsilon' \gg 1$)	Units
$\alpha =$	$\omega \left[\frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right]^{1/2}$	0	$\frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\pi f \mu \sigma}$	(Np/m)
$\beta =$	$\omega \left[\frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right]^{1/2}$	$\omega \sqrt{\mu \epsilon}$	$\omega \sqrt{\mu \epsilon}$	$\sqrt{\pi f \mu \sigma}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu}{\epsilon'}} \left(1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2}$	$\sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\frac{\mu}{\epsilon}}$	$(1 + j) \frac{\alpha}{\sigma}$	(Ω)
$u_p =$	ω / β	$1 / \sqrt{\mu \epsilon}$	$1 / \sqrt{\mu \epsilon}$	$\sqrt{4\pi f / \mu \sigma}$	(m/s)
$\lambda =$	$2\pi / \beta = u_p / f$	u_p / f	u_p / f	u_p / f	(m)
Notes: $\epsilon' = \epsilon$; $\epsilon'' = \sigma / \omega$; in free space, $\epsilon = \epsilon_0$, $\mu = \mu_0$; in practice, a material is considered a low-loss medium if $\epsilon''/\epsilon' = \sigma / \omega \epsilon < 0.1$ and a good conducting medium if $\epsilon''/\epsilon' > 10$.					

$|k_c| = \omega \sqrt{\mu \epsilon_c} = \omega \sqrt{\mu \epsilon_0 \epsilon_{rc}} = \omega \sqrt{\mu \epsilon_0 (\epsilon_r' - j \epsilon_r'')} = \beta - j \alpha$

- if does not fit in neither low-loss nor good conductor use above formula a whack calculator, don't use the one in the table

$\delta = \text{skin depth} = \text{depth at which attenuation is } e^{-1} = \frac{1}{\alpha}$

Good Conductor: $\eta_c = (1 + j) \frac{\alpha}{\sigma} = \sqrt{2} \frac{\alpha}{\sigma} e^{j\pi/4}$

$S(r, t) = E(r, t) \times H(r, t)$

$S(r, t) = \text{Re}\{E(r) e^{j\omega t}\} \times \text{Re}\{H(r) e^{j\omega t}\}$

- This is cross product between two vectors

$S_{ave} = \frac{1}{2} \text{Re}\{E(r) \times H(r)^*\}$

- this is multiplication of E phasor with conjugate of H phasor

$$S_{ave} = \frac{1}{2} Re\left\{\frac{1}{\eta_c}\right\} |E_0|^2 e^{-2\alpha(a_k \cdot r)} a_k$$

$$S_{ave} = \frac{1}{2} Re\{\eta_c\} |H_0|^2 e^{-2\alpha(a_k \cdot r)} a_k$$

$$S_{ave} = P_{ave} a_k$$

- all in W/m²
- S_{ave} also called power density

Summary: Reflection and Transmission

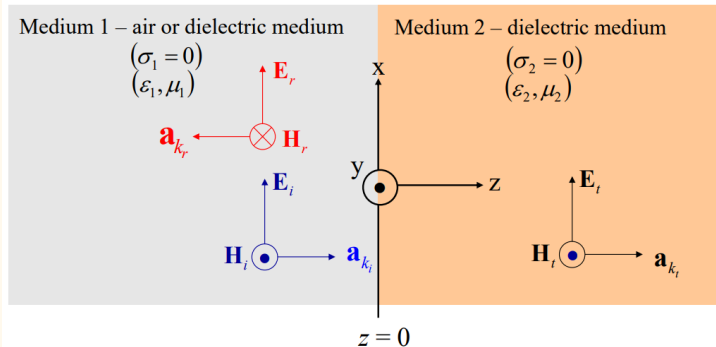
<https://www.desmos.com/calculator/n199kxhx72>

At the boundary of two dielectric media:
2 equations are considered in solving the reflected and transmitted waves

Table 6-2: Boundary conditions for the electric and magnetic fields.

Field Components	General Form	Medium 1 Dielectric	Medium 2 Dielectric	Medium 1 Dielectric	Medium 2 Conductor
Tangential E	$\hat{n}_2 \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$	$E_{1t} = E_{2t}$		$E_{1t} = E_{2t} = 0$	
Normal D	$\hat{n}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$	$D_{1n} - D_{2n} = \rho_s$		$D_{1n} = \rho_s$	$D_{2n} = 0$
Tangential H	$\hat{n}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$	$H_{1t} = H_{2t}$		$H_{1t} = J_s$	$H_{2t} = 0$
Normal B	$\hat{n}_2 \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0$	$B_{1n} = B_{2n}$		$B_{1n} = B_{2n} = 0$	

Notes: (1) ρ_s is the surface charge density at the boundary; (2) $\mathbf{J}_s$ is the surface current density at the boundary; (3) normal components of all fields are along $\hat{n}_2$, the outward unit vector of medium 2; (4) $E_{1t} = E_{2t}$ implies that the tangential components are equal in magnitude and parallel in direction; (5) direction of $\mathbf{J}_s$ is orthogonal to $(\mathbf{H}_1 - \mathbf{H}_2)$.



Medium 1 and 2 are lossless (dielectric) mediums

Reflection coefficient

$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

Transmission coefficient

$$\tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$1 + \Gamma = \tau \quad |\Gamma| \leq 1$$

Incident, Reflected and Transmitted E-field

Incident, Reflected and Transmitted H-field

$E_i(z) = a_x E_{i0} e^{-j\beta_1 z}$ $E_r(z) = a_x E_{r0} e^{j\beta_1 z} = a_x \Gamma E_{i0} e^{j\beta_1 z}$ $E_t(z) = a_x E_{t0} e^{-j\beta_2 z} = a_x \tau E_{i0} e^{-j\beta_2 z}$		$H_i(z) = a_y H_{i0} e^{-j\beta_1 z} = a_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z}$ $H_r(z) = -a_y H_{r0} e^{j\beta_1 z} = -a_y \frac{E_{r0}}{\eta_1} e^{j\beta_1 z}$ $H_r(z) = -a_y \frac{\Gamma E_{i0}}{\eta_1} e^{j\beta_1 z}$ $H_t(z) = a_y H_{t0} e^{-j\beta_2 z} = a_y \frac{E_{t0}}{\eta_2} e^{-j\beta_2 z}$ $H_t(z) = a_y \frac{\tau E_{i0}}{\eta_2} e^{-j\beta_2 z}$	
<u>Reflected Power Density, S_r</u>	<u>Transmitted Power Density, S_t</u>	<u>Conservation of Energy</u> Question about intensity of EM waves - Physics Stack Exchange	
$S_r = \frac{1}{2} \text{Re}\{E_r(z) \times H_r(z)^*\}$ $S_r = \frac{1}{2} \text{Re}\{(-a_z) \Gamma ^2 \frac{ E_{i0} ^2}{\eta_1}\}$ $S_r = (-a_z) \frac{1}{2} \Gamma ^2 \frac{ E_{i0} ^2}{\eta_1}$ $S_r = (-a_z) \Gamma ^2 S_i $ $ \Gamma ^2 = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}\right)^2 = \frac{ \text{reflected power density} }{ \text{incident power density} }$ all in W/m²	$S_t = \frac{1}{2} \text{Re}\{E_t(z) \times H_t(z)^*\}$ $S_t = \frac{1}{2} \text{Re}\{a_z \tau ^2 \frac{ E_{i0} ^2}{\eta_2}\}$ $S_t = a_z \tau ^2 \frac{1}{2} \frac{\eta_1}{\eta_2} \frac{ E_{i0} ^2}{\eta_1}$ $S_t = a_z \tau ^2 \frac{\eta_1}{\eta_2} S_i $ $ \tau ^2 \frac{\eta_1}{\eta_2} = \left(\frac{2\eta_2}{\eta_2 + \eta_1}\right)^2 \frac{\eta_1}{\eta_2} = \frac{4\eta_1 \eta_2}{(\eta_2 + \eta_1)^2}$ $= \frac{ \text{transmitted power density} }{ \text{incident power density} }$ all in W/m²	$ \Gamma ^2 + \tau ^2 \frac{\eta_1}{\eta_2} = 1$ $S_1 = \text{total ave power density in } m_1$ $S_1 = \frac{1}{2} \text{Re}\{E_1(z) \times H_1(z)^*\}$ $S_1 = \frac{1}{2} \text{Re}\{[E_i(z) + E_r(z)] \times [H_i(z)^* + H_r(z)^*]\}$ $S_1 = \frac{1}{2} \text{Re}\{E_i(z) \times H_i(z)^*\} + \frac{1}{2} \text{Re}\{E_r(z) \times H_r(z)^*\}$ $S_1 = a_z (1 - \Gamma ^2) S_i $ $S_1 = S_t$ Movement of power in m2 must be from the overall movement in m1 For perfect conductor m2: $S_1 = a_z (1 - -1 ^2) S_i = 0$	
<u>Total E-field in Medium 1 (Standing Wave with Non-zero Node)</u>		<u>Total H-field in Medium 1 (Standing Wave with Non-zero Node)</u>	

$$E_1(z) = E_i(z) + E_r(z)$$

$$E_1(z) = a_x E_{i0} e^{-j\beta_1 z} + a_x \Gamma E_{i0} e^{j\beta_1 z}$$

$$E_1(z) = a_x E_{i0} e^{-j\beta_1 z} (1 + \Gamma e^{j2\beta_1 z})$$

$$\text{let } \Gamma = |\Gamma|e^{j\theta} \text{ \& } z' = -z$$

$$E_1(z) = a_x E_{i0} e^{j\beta_1 z'} (1 + |\Gamma|e^{j\theta} e^{-j2\beta_1 z'})$$

$$E_1(z) = a_x E_{i0} e^{j\beta_1 z'} (|\Gamma|e^{j\theta-2j\beta_1 z'} - (-1))$$

$$|E_1(z)| = |E_{i0}| | |\Gamma|e^{j(\theta-2\beta_1 z')} - (-1) |$$

$$H_1(z) = H_i(z) + H_r(z)$$

$$H_1(z) = a_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} + -a_y \frac{\Gamma E_{i0}}{\eta_1} e^{j\beta_1 z}$$

$$H_1(z) = a_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} (1 - \Gamma e^{j2\beta_1 z})$$

$$\text{let } \Gamma = |\Gamma|e^{j\theta} \text{ \& } z' = -z$$

$$H_1(z) = a_y \frac{E_{i0}}{\eta_1} e^{j\beta_1 z'} (1 - |\Gamma|e^{j\theta} e^{-j2\beta_1 z'})$$

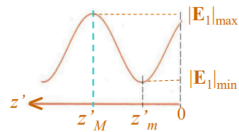
$$|H_1(z)| = \frac{|E_{i0}|}{\eta_1} |1 - |\Gamma|e^{j(\theta-2\beta_1 z')}|$$

	$\Gamma > 0 \quad (\eta_2 > \eta_1)$	$\Gamma < 0 \quad (\eta_2 < \eta_1)$
$ E_1 _{\max}, H_1 _{\min}$	$ E_{i0} (1+ \Gamma), \frac{ E_{i0} }{\eta_1}(1- \Gamma)$	$ E_{i0} (1+ \Gamma), \frac{ E_{i0} }{\eta_1}(1- \Gamma)$
Condition	$2\beta_1 z'_M = 2n\pi$	$2\beta_1 z'_M = (2n+1)\pi$
Position	$z'_M = n\frac{\lambda_1}{2}, n = 0, 1, 2, \dots$	$z'_M = \frac{\lambda_1}{4} + n\frac{\lambda_1}{2}, n = 0, 1, 2, \dots$
$ E_1 _{\min}, H_1 _{\max}$	$ E_{i0} (1- \Gamma), \frac{ E_{i0} }{\eta_1}(1+ \Gamma)$	$ E_{i0} (1- \Gamma), \frac{ E_{i0} }{\eta_1}(1+ \Gamma)$
Condition	$2\beta_1 z'_m = (2n+1)\pi$	$2\beta_1 z'_m = 2n\pi$
Position	$z'_m = \frac{\lambda_1}{4} + n\frac{\lambda_1}{2}, n = 0, 1, 2, \dots$	$z'_m = n\frac{\lambda_1}{2}, n = 0, 1, 2, \dots$

Attention

Note that: $|E_1|_{\max} = |E_{i0}|(1+|\Gamma|)$

$|E_1|_{\min} = |E_{i0}|(1-|\Gamma|)$



(1) H_1 's maxima and minima are opposite to those of E_1 's.

(2) Since $\beta_1 \lambda = 2\pi$, when z' increases by λ , $|H_1(z')|$ (as well as $|E_1(z')|$) experiences 2 periods since $2\beta_1 z'$ is in the exponent.

- Position of maxima and minima are spaced out every half wavelength
- With lossless media, $\Gamma = |\Gamma|e^{j\theta}$, Γ is real, hence, θ is either 0 or $-\pi$ and position of maxima and minima is simplified to the table to the left
- Table to the left gives value in z' , multiply -ve to get back to the original z axis

Standing Wave Ratio

The ratio of $|E_1|_{\max}$ to $|E_1|_{\min}$ is called

the **standing wave ratio (SWR) S**:

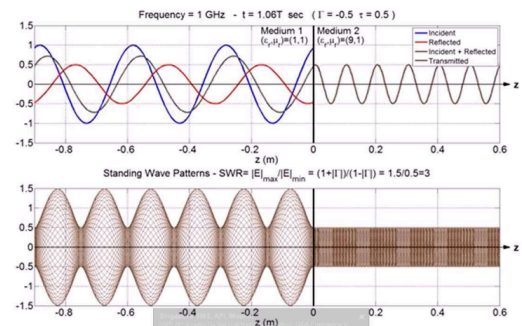
$$S = \frac{|E_1|_{\max}}{|E_1|_{\min}} = \frac{1+|\Gamma|}{1-|\Gamma|}$$

It is easy to find

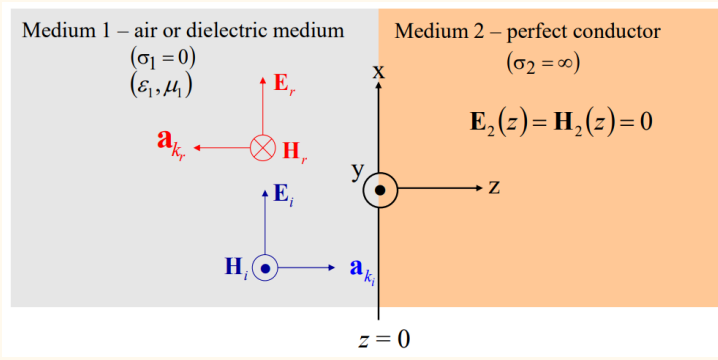
$$|\Gamma| = \frac{S-1}{S+1}$$

- SWR = 1 if impedance is matched
- SWR > 0 if impedance is mismatch
- SWR = ∞ if $z_2 = 0$

Wave reflection and transmission at boundary of two dielectric media



Special Case: Medium 2 is perfect conductor (all metal, sea water need to check)



Perfect conductor: $\eta_2 \rightarrow 0$

Reflection coefficient

$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -1$$

Transmission coefficient

$$\tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1} = 0$$

In perfect conductor, E and H attenuates at skin depth = 0, hence E and H = 0 in perfect conductor.

Incident, Reflected and Transmitted E-field

$$E_i(z) = a_x E_{i0} e^{-j\beta_1 z}$$

$$E_r(z) = a_x E_{r0} e^{j\beta_1 z} = -a_x E_{i0} e^{j\beta_1 z}$$

$$E_t(z) = 0$$

Incident, Reflected and Transmitted H-field

$$H_i(z) = a_y H_{i0} e^{-j\beta_1 z} = a_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z}$$

$$H_r(z) = -a_y H_{r0} e^{j\beta_1 z} = -a_y \frac{E_{r0}}{\eta_1} e^{j\beta_1 z}$$

$$H_t(z) = a_y \frac{E_{i0}}{\eta_1} e^{j\beta_1 z}$$

$$H_t(z) = 0$$

Total E-field in Medium 1 (Standing Wave with Zero Node)

Phasor

$$E_1(z) = E_i(z) + E_r(z)$$

$$E_1(z) = a_x E_{i0} e^{-j\beta_1 z} - a_x E_{i0} e^{j\beta_1 z}$$

$$E_1(z) = a_x E_{i0} (e^{-j\beta_1 z} - e^{j\beta_1 z})$$

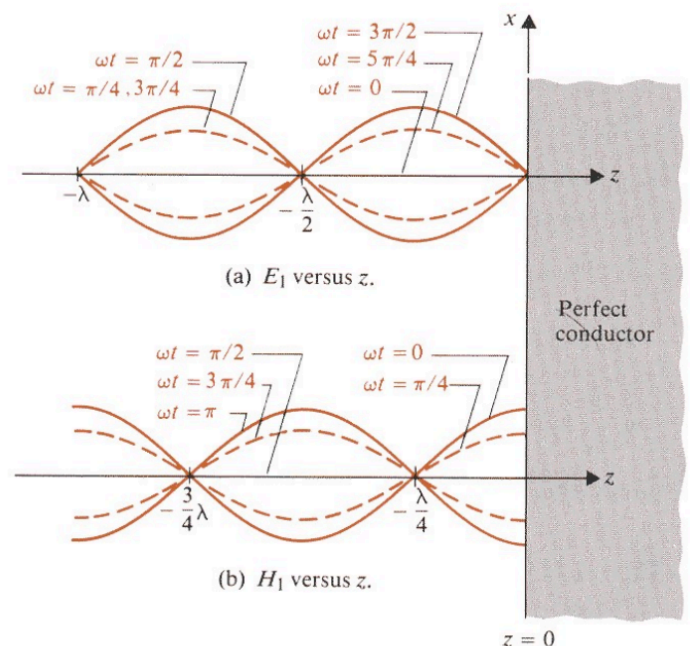
$$E_1(z) = -a_x j 2 E_{i0} \sin(\beta_1 z)$$

Time Domain (Instantaneous)

$$E_1(z, t) = \text{Re}\{E_1(z) e^{j\omega t}\}$$

$$E_1(z) = a_x 2 E_{i0} \sin(\beta_1 z) \sin(\omega t)$$

Total H-field in Medium 1 (Standing Wave with Zero Node)



Total electric and magnetic fields in medium 1

- E field at metal surface is always 0
- H field at metal surface is always a maxima

Phasor

$$H_1(z) = H_i(z) + H_r(z)$$

$$H_1(z) = a_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} + a_y \frac{E_{i0}}{\eta_1} e^{j\beta_1 z}$$

$$H_1(z) = a_y \frac{E_{i0}}{\eta_1} (e^{-j\beta_1 z} + e^{j\beta_1 z})$$

$$H_1(z) = a_y \frac{E_{i0}}{\eta_1} 2\cos(\beta_1 z)$$

Time Domain (Instantaneous)

$$H_1(z, t) = \text{Re}\{H_1(z)e^{j\omega t}\}$$

$$E_1(z) = a_y 2 \frac{E_{i0}}{\eta_1} \cos(\beta_1 z) \cos(\omega t)$$

Note that both the total electric and total magnetic fields in medium 1 are **standing waves**.

By setting $\sin(\beta_1 z) = 0$ in E_1

1. They are $\perp$ to each other and 90° out of phase.

2. The electric field vanishes at $z = -n\lambda/2, n = 0, 1, 2, \dots$

3. The magnetic field vanishes at $z = -(\lambda/4 + n\lambda/2)$.

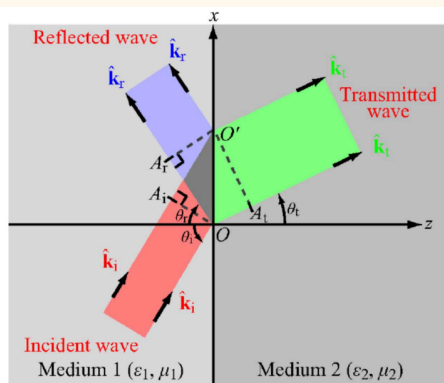
Negative sign:
Medium 1 is on the left of origin: $z < 0$

By setting $\cos(\beta_1 z) = 0$ in H_1

$\Gamma < 0 \quad (\eta_2 < \eta_1)$
$ E_{i0} (1 + \Gamma), \frac{ E_{i0} }{\eta_1}(1 - \Gamma)$
$2\beta_1 z'_M = (2n + 1)\pi$
$z'_M = \frac{\lambda_1}{4} + n \frac{\lambda_1}{2}, n = 0, 1, 2, \dots$
$ E_{i0} (1 - \Gamma), \frac{ E_{i0} }{\eta_1}(1 + \Gamma)$
$2\beta_1 z'_m = 2n\pi$
$z'_m = n \frac{\lambda_1}{2}, n = 0, 1, 2, \dots$

this is just the case of normal case but $\Gamma < 0$

Oblique Incident Wave



Snell's Law:

$$\theta_r = \theta_i$$

$$n_2 \sin \theta_t = n_1 \sin \theta_i$$

Refractive index:

$$n = \sqrt{\epsilon_r \mu_r}$$